Subconscious: Format-Matched Memory as Cognitive Architecture
for Large Language Models on Heterogeneous Consumer Silicon

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Abstract

A 39-entry key-value lookup table resolved factual recall failures that a full multi-signal retrieval pipeline with 7,395 memory records could not. The model did not change. The retrieval algorithm did not change. The storage format changed. This finding motivates the central claim: the five major failure modes of deployed LLMs (hallucination, staleness, contradiction, depersonalization, and context loss) are not model deficiencies but format mismatches, and each maps to a specific, empirically measured weakness of the base model. We present Subconscious, a cognitive architecture that functions as a prosthetic for these measured deficiencies. Seven format-matched stores compensate for five model weaknesses, running as a continuous background process on heterogeneous Apple Silicon, with real-time perception on dedicated neural silicon concurrent with generation on GPU at zero measured contention. The system extracts typed perceptions (facts, speculations, corrections, decisions) from live conversation, captures not just what the user knows but how they reason, what they have tried and abandoned, and how their understanding has evolved. The user's data, decisions, and patterns of thinking shape the system's retrieval, consolidation, and response behavior. Over time, the architecture adapts to its user through continuous background consolidation: an always-on process that digests, integrates, and prepares without human intervention, so that each session opens with the accumulated understanding of every session before it. Evaluated across 20 queries against a raw Qwen2.5-72B baseline, the system eliminates acronym hallucination (0/4 to 4/4), enables cross-session arc synthesis (0/4 to 4/4), and surfaces dead-path corrections proactively. A self-reinforcing hallucination loop, where the system retrieved its own prior fabrications as authoritative evidence, was discovered and resolved through provenance-weighted retrieval. All code is open source under MIT license.

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1. Introduction

Asked "What is the SLC geometry of the M5 Pro?", a Qwen2.5-72B model responds: "The SLC (Single Layer Capacitor) geometry..." It does not know that SLC, in the context of Apple Silicon research, refers to the System Level Cache, a ~48MB shared cache with 21-way associativity and 128-byte lines, measured through months of reverse engineering on the hardware in question. The model hallucinates a plausible expansion of the acronym from its training distribution and proceeds with confidence.

Asked the same question with Subconscious active, the system responds: "21-way, with 128B lines, totaling approximately 48MB." The answer is correct, specific, and includes provenance. The model weights did not change. The architecture around the model changed. The system retrieved the canonical measurement from a format-matched store and injected it as context. The model's job shifted from generating a plausible answer to reasoning over a correct one.

This paper argues that the five most common failure modes of deployed LLMs are not model deficiencies but architectural absences. They arise because the model operates without the memory infrastructure that any functioning cognitive system requires. We demonstrate that format-matched memory, where different types of knowledge are stored in formats optimized for their retrieval patterns, resolves all five failure modes from a single design principle. The contribution is not a better retrieval algorithm. It is the observation that retrieval quality is determined by storage format, and that different knowledge types demand different formats.

2. Five Failure Modes, One Cause

We identify five systematic failure modes in LLMs deployed without persistent memory. Each is conventionally treated as a separate research problem. We argue they share a single cause: the absence of format-matched memory stores.

2.1 Hallucination as Storage Absence

When a model encounters a domain-specific term not well-represented in its training data, it generates a plausible completion. 'SLC' becomes 'Single Layer Capacitor.' 'ANE' becomes 'Adobe Native Extension.' These are not reasoning failures. They are the expected behavior of a system that must generate when it cannot recall. Every hallucination of domain-specific knowledge is a missing record in the right store. The model fills the gap because no external store provides the answer.

2.2 Staleness as Format Mismatch

Training cutoffs freeze model knowledge at a point in time. RAG is the standard remedy, but RAG treats all updated knowledge uniformly: embed it and retrieve by similarity. This works for narrative knowledge ('what happened recently') but fails for point-value knowledge. If a measurement was updated from 200 to 307, an embedding store contains both values as semantically similar records, producing retrieval interference. A measurement registry keyed by entity name contains only the canonical value. Staleness is solved not by adding new data, but by adding it in a format where the current value is distinguishable from the historical value.

2.3 Contradiction as Missing Index

Models contradict themselves across conversations because they have no mechanism to detect that a new statement conflicts with a prior one. A correction store with entity-indexed contradiction detection reduces this to a lookup rather than a reasoning task. The model does not need to 'remember' that a research path was killed. The correction store surfaces the evidence automatically when the relevant entity appears in a new query.

2.4 Depersonalization as Missing Ambient Context

Every conversation starts from the same generic model. A profile store with pre-loaded context solves this through ambient retrieval: the profile is injected into every interaction without being queried. The user's methodology preferences, decision patterns, and standing rules shape every response. Personalization is not a model capability problem. It is the absence of a storage format optimized for always-present injection.

2.5 Context Loss as Missing Session State

'I don\'t have any previous conversation.' A session store with recency retrieval solves this: the bridge block, the warm open, the last session's state. Context loss persists not because context windows are too small, but because the previous session's state is not stored in a format that makes it injectable at session open.

3. Format-Matched Memory

The central claim of this paper is that memory systems fail when they treat knowledge as uniform. Different knowledge types have different intrinsic retrieval shapes, and those shapes demand matched storage formats, retrieval mechanisms, and consolidation strategies.

The observation that structured queries require structured storage is not novel in isolation. Any database engineer knows that key-value lookups outperform similarity search for exact values. The contribution is the unified framework: that all five failure modes of deployed LLMs trace to format mismatches, that the format-retrieval-consolidation triple must be internally consistent per knowledge type, and that a continuous background process maintaining multiple format-matched stores resolves these failures from a single architectural principle while adapting to the user's evolving knowledge and cognitive patterns. The individual stores are simple. The principle that connects them, and the continuous process that maintains them from the user's own data, is the contribution.

Table 1. *Seven knowledge types and their format-matched stores.*

Knowledge Type	Retrieval Shape	Storage Format	Retrieval Mechanism	Consolidation
Narrative (arcs, corrections, decisions)	Path: entity through time	Timestamped records with entity tags and categories	Entity threading + arc tracing	Semantic supersession
Measurement (point values)	Point: entity to value	Key-value pairs with units and provenance	Direct entity lookup	Correction (same entity only)
Profile (preferences, patterns)	Ambient: always present	Compiled summaries	Pre-injection at session open	Periodic recompilation
Session state (continuity)	Recency: most recent	Last-session snapshot	Load most recent on open	Replacement
Corrections (dead paths)	Adversarial: check against input	Entity-indexed contradiction records	Entity match against user statements	Append-only
Collections (enumerable sets)	Enumeration: filter by tag	Tag-indexed record sets	Tag lookup, no truncation	Tag management
Unknown knowledge (gaps)	Absence: detect and guard	Domain keyword index	Topical relevance check + guard injection	Guard when irrelevant

The format-retrieval-consolidation triple must be internally consistent for each knowledge type. Narrative knowledge consolidates by supersession (newer findings retire older ones). Measurements consolidate by correction (only a newer measurement of the same entity supersedes). Profiles consolidate by recompilation. Session state consolidates by replacement. Applying the wrong consolidation strategy to a knowledge type destroys information the retrieval path depends on. Superseding a measurement because a semantically similar but different measurement was stored more recently is the geometric interference problem identified by Gopinath

(2026) operating at the application layer.

4. Architecture

4.1 Subconscious as Continuous Process

Subconscious is not a memory system that activates on query. It is a continuous cognitive process that runs perception, integration, consolidation, and recall preparation simultaneously with conscious generation, at all times, whether the user is in session or not.

During a session, while the primary model generates a response, the extraction model simultaneously processes every message on dedicated neural silicon, classifying each statement into eight epistemological categories: fact, speculation, correction, decision, preference, plan, question, and goal. In measured production use, the extraction model fires within 4 seconds of generation start, and the resulting typed perceptions are embedded and queryable within 60 seconds. This is not post-processing. This is perception happening concurrently with cognition, enabled by the physical independence of the hardware paths.

The epistemological categories capture not just what the user knows, but how the user reasons. A speculation followed by a measurement followed by a correction is a reasoning trace: the user hypothesized, tested, and updated. A decision record captures not just the conclusion but, via the thread of records that preceded it, the chain of evidence that led to it. When the system later retrieves a decision, the arc tracer reconstructs this provenance: 'Initially speculated X, measured Y which contradicted, corrected to Z, decided W.' The user's cognitive style, their patterns of inquiry, their willingness to abandon failed hypotheses, their preference for measurement over argument, all emerge from the category sequences across sessions and are compiled into the profile that shapes future retrieval and response behavior.

Provenance is therefore not a metadata annotation but an architectural invariant of every store. Every record carries an explicit source_role (user, canonical, assistant, derived) and retrieval cannot treat these roles as interchangeable. A user-stated measurement and a model-generated rephrasing of that measurement must not score identically at recall time, because conflating them lets the system re-ingest its own prior output and treat it as ground truth on the next query. This is not a hypothetical risk. We observed it in production (Section 6.5): of 7,640 stored memories, 2,668 (35 percent) had source_role 'assistant', and without provenance weighting the system retrieved its own past fabrications at the same rank as user-provided facts, generating new responses grounded in them, which were then extracted and stored again. The architectural fix is that assistant-sourced records are scored with a 0.5x multiplier and the measurement registry, canonical state store, and correction store accept writes only from user or canonical sources. Provenance constrains what the system is allowed to learn from itself. A cognitive architecture that stores model output alongside user input without this constraint is structurally unstable: it will drift toward its own priors over time, with no mechanism to recover.

Between sessions, nine maintenance loops run on a scheduled cycle: semantic supersession retires near-duplicate records, contradiction resolution integrates conflicting findings, the user profile recompiles from the growing corpus, the entity index rebuilds, and narrative threads update. This is the process that makes the system adaptive. Each maintenance cycle incorporates the latest session's perceptions into the long-term knowledge structure. The profile evolves. The entity vocabulary expands. The correction chains lengthen. The measurement registry updates with newly extracted canonical values. When the next session opens, the system reflects everything it has learned, not through explicit programming but through continuous consolidation of the user's own cognitive output.

The biological analogy is not sleep consolidation, which implies discrete phases. The analogy is the actual subconscious: the continuous process that runs underneath conscious attention, integrating experience in real time, updating models of the world, and preparing retrieval structures for the next conscious query. The subconscious does not have an off state.

4.2 Hardware Separation as Functional Necessity

The heterogeneous hardware mapping is not a performance optimization. It is what makes concurrent perception possible at all. If the extraction model ran on the same GPU as the generation model, it would either contend with generation (degrading the user experience) or wait until generation completes (making extraction post-hoc, not real-time). Either outcome collapses the architecture to sequential processing: generate, then extract, then consolidate. That is batch processing, not a subconscious.

The ANE with its independent DMA channel (111 GB/s, measured as physically separate from the GPU's 241 GB/s sustained bandwidth) is what makes the 4-second extraction-after-generation timing possible. The +2.4% contention measurement is not a benchmark result. It is proof that the cognitive architecture is physically realizable: two cognitive processes running simultaneously on independent silicon without interference.

Table 2. *Cognitive function to hardware mapping.*

Cognitive Function	Hardware	Bandwidth	Measured Contention
Conscious generation (72B)	GPU (20 cores)	241 GB/s sustained	Baseline
Subliminal perception (8B)	ANE (16-core tile array)	111 GB/s dedicated DMA	+2.4% (noise)
Routing (80M classifier)	ANE SRAM	On-chip, no DRAM	905 microseconds
Retrieval (embedding search)	CPU/AMX	Shared DRAM	0.84ms per embed
Consolidation (maintenance)	CPU	Shared DRAM	Background, hourly

The mapping is not arbitrary. We formalize each cognitive function as a requirements tuple (bandwidth, compute intensity, latency constraint, power budget, concurrency requirement, access pattern) and find that the contention-free assignment to available accelerators is the only one satisfying all constraints simultaneously when conscious generation and subliminal perception must run concurrently. GPU-only deployment requires 91.5% bus utilization for generation plus perception, exceeding stability thresholds. Heterogeneous deployment distributes this to 78% GPU utilization plus 36% on the ANE's independent DMA channel.

4.3 Designing Around Measured Model Capabilities

The architecture was not derived from a theoretical taxonomy and then validated. It was designed from empirical observation of where the base model fails, what kind of context makes it succeed, and how the user's own data should shape the system's behavior. Each store exists because a specific failure mode was observed in production, its root cause was diagnosed as a context-format problem, and a format-matched solution was built and validated.

The measurement registry exists because the 72B was observed returning 256 GB/s when the correct answer was 307 GB/s. The root cause: semantically similar measurements interfere in embedding space. The fix: direct entity lookup that bypasses embedding entirely. The correction store exists because the 72B was observed affirming dead research paths. The root cause: no mechanism to check new statements against prior corrections. The fix: entity-indexed contradiction records scanned on every query. The narrative store exists because top-15 memory snippets produced fragmented responses while chronological arcs produced coherent ones. The root cause: the 72B reasons well over structured narratives but poorly over disconnected fragments. The fix: thread detection and arc tracing that present evolution rather than snapshots.

This empirical design process means the architecture is fitted to the model's cognitive profile, not to one user's domain. The 72B hallucinating domain acronyms is not specific to hardware research. Any 72B-class model will hallucinate domain terms absent from its training data, confuse semantically similar point values, and fail to maintain cross-session state. A lawyer's model will hallucinate case citations. A physician's model will confuse similar drug dosages. The seven stores address five model weaknesses that are model-intrinsic, not domain-specific. A different user in a different domain, using the same class of model, would benefit from the same seven stores.

What makes the system user-specific is not the architecture but the content. The extraction pipeline populates the stores from the user's conversations. The entity vocabulary reflects the user's domain. The correction chains reflect the user's evolving understanding. The profile reflects the user's cognitive patterns. The measurement registry reflects the user's canonical values. The architecture is the skeleton. The user's data is the organism that grows on it. Over weeks of natural use, the system increasingly reflects how this particular user thinks, what they have tried and abandoned, what they consider settled, and how they prefer information to be presented. This adaptation requires no configuration, no tuning, and no manual curation. The user's own cognitive output, captured in real time by the perception layer, is the input that shapes the system.

5. Implementation

5.1 Production Stack

Table 3. *Production inference components.*

Component	Model	Hardware	Throughput
Conscious generation	Qwen2.5-72B-Q4	GPU	6.24 cold / 29.3 warm tok/s
Subliminal perception	Llama-3.1-8B-Q8	ANE	7.9 tok/s
N-gram drafter	N-gram table	CPU	91% draft rate
Embedding	MiniLM CoreML	ANE	0.84ms/embed
Router	80M classifier	ANE SRAM	905 microseconds

5.2 Retrieval Pipeline

Retrieval operates in three stages. First, the intent filter classifies the incoming query as requiring arc retrieval (narrative), point retrieval (measurement), or default retrieval (top-k snippets). Intent classification uses surface signals from the query: past-tense verbs and evolution keywords route to narrative; entity-value question patterns route to measurement; unmatched queries fall through to default.

For narrative queries, the thread detector extracts entities from the query, looks up the entity index (316 named entities, 7,395 records across 104 sessions), and retrieves all records sharing those entities, ordered chronologically. The arc tracer classifies the thread shape (reversal, convergence, dead-end) from the category sequence. The narrative packager produces a structured context block injected into the generation model's prompt. Total latency: 8-23ms.

For measurement queries, the registry performs a direct entity lookup against 39 canonical entries with provenance metadata. No embedding search, no ranking, no interpretation. Total latency: sub-millisecond.

Full pipeline overhead per query: intent classification (<1ms) + retrieval (sub-ms for registry, 8-23ms for narrative, 15-30ms for default) + contradiction check (<5ms) + profile injection (pre-loaded, 0ms) + session bridge (pre-loaded, 0ms). Total: 5-30ms depending on retrieval path. This represents less than 0.2% of typical 72B generation time at 17.5 tok/s. The memory infrastructure is invisible in the user's experience of response latency.

5.3 Multi-Signal Retrieval

Default retrieval uses five-signal fusion: embedding similarity (weight 0.35), entity match (0.25), type match (0.15), impact chain (0.15), and temporal recency (0.10). This composite retrieval expands the effective dimensionality of the retrieval space beyond what the embedding model provides. MiniLM, with a nominal dimensionality of 384, concentrates its variance in approximately 16 effective dimensions (Gopinath, 2026). Entity match adds categorical dimensions that do not suffer from angular interference. In empirical testing at 5,647 memories, composite retrieval achieves 100% Hit@5 on queries where the answer exists in the store, compared to 92% for embedding-only retrieval. Composite retrieval is a strict superset: it never loses a hit that embedding alone finds.

5.4 Consolidation as Interference Management

Gopinath (2026) demonstrates that forgetting in embedding-based memory systems follows the Ebbinghaus power law, driven by interference from competing memories rather than temporal decay. The forgetting exponent b increases monotonically with competitor count. Semantic supersession directly manages this: by retiring near-duplicate records (1,568 superseded to date), the system reduces the competitor count in the embedding space. Critically, targeted supersession of semantically similar records provides 5-10x more interference relief than random removal, because it eliminates the memories that cluster most tightly in the low-dimensional effective space where interference concentrates.

Under continuous consolidation with targeted supersession, our analysis indicates that the forgetting exponent b is bounded above by b(n_irreducible), where n_irreducible is the number of genuinely distinct memories that cannot be further consolidated. Without consolidation, b grows without bound as the memory store grows. This is the qualitative difference between a managed and an unmanaged memory store: bounded versus unbounded forgetting.

6. Evaluation

6.1 Experimental Setup

We evaluate across 20 queries in six categories: factual recall (4), arc synthesis (4), proactive correction (4), cross-domain (2), cold baseline (2), and warm session open (4). Each query is run under two conditions: the raw Qwen2.5-72B with no memory infrastructure (baseline), and the full Subconscious system with all seven stores active. The corpus spans 104 sessions (44 Claude.ai conversations + 60 Claude Code sessions) containing 7,395 extracted records over 5 weeks of sustained technical research.

6.2 Results

Table 4. *Phase 5 evaluation results by category.*

Category	Baseline (raw 72B)	Full System	Store Responsible
Factual recall (4 queries)	0/4. Hallucinated acronyms (SLC = 'Single Layer Capacitor', ANE = 'Adobe Native Extension'), fabricated values (89 GB/s for DRAM)	4/4. Correct canonical values with source provenance	Measurement registry
Arc synthesis (4 queries)	0/4. 'There might be some context missing', 'not a standard or widely recognized term'	4/4. Cross-session evolution traces with specific milestones and dates	Narrative store + thread detector
Proactive correction (4 queries)	0/4. Generic descriptions, affirmed dead paths ('Yes, cache hints can help')	3/4. Dead paths surfaced with evidence. One soft hedge instead of firm rejection	Correction store
Cold baseline (2 queries)	2/2. Reasonable textbook explanations	2/2. Equivalent quality	None (no memory advantage expected)
Cross-domain (2 queries)	0/2. No awareness of connections between research areas	2/2. Links ANE hardware findings to cognitive architecture decisions	Narrative store + profile store
Warm open (4 queries)	0/4. 'I don\'t have any previous conversation'	4/4. Continues from project state, suggests specific next steps	Session store + profile store

The system produces categorical improvement in four of six categories. Cold baseline queries (general knowledge not requiring memory) show no degradation, confirming that the memory infrastructure does not interfere with the model's native capabilities. The single proactive correction miss (soft hedge instead of firm rejection) reflects generation variance: the correction record was present in context, but the model chose a softer phrasing. The knowledge was retrieved correctly; the generation was stochastic.

6.3 Format as Architecture: The Registry Experiment

The most informative result is the progression of factual recall quality across three system configurations:

Table 5. *Factual recall across system configurations.*

Configuration	Factual Recall	Example (DRAM bandwidth query)
Raw 72B (no memory)	0/4	'Approximately 89 GB/s' (hallucinated)
Subconscious v1 (narrative + profile + session + correction stores)	1/4	'256 GB/s' (wrong: retrieved SLC bandwidth, not DRAM)
Subconscious v2 (add 39-entry measurement registry)	4/4	'307 GB/s' (correct, with source provenance)

The transition from v1 to v2 is the paper's central empirical finding. The full retrieval pipeline with 7,395 memory records, five-signal fusion, entity threading, and arc tracing could not correctly recall that DRAM bandwidth is 307 GB/s. A 39-entry JSON lookup table could. The model did not change. The retrieval algorithm did not change. The embedding model did not change. A flat key-value map resolved what the entire multi-signal retrieval infrastructure could not, because the storage format matched the retrieval need.

The v1 failure is precisely characterizable: 'DRAM bandwidth' and 'SLC bandwidth' and 'AMCC ceiling' are semantically similar in embedding space (all are bandwidth measurements of the same machine). The five-signal retrieval correctly found bandwidth-related records but could not discriminate between them. The 72B, given multiple similar measurements in its context window, selected the wrong one. This is the DRM false memory effect (Deese, 1959; Roediger & McDermott, 1995) operating in a production retrieval system: the 'lure' (256 GB/s SLC bandwidth) is geometrically close to the 'target' (307 GB/s DRAM bandwidth), and the system recalls the wrong one.

The measurement registry eliminates this interference by removing the query from the embedding space entirely. A direct lookup by entity name has no geometric interference because it operates in a categorical space, not a vector space. The failure was not a limitation of the retrieval algorithm. It was a property of the storage format.

6.4 Self-Correction in Live Use

In live interactive testing, a three-turn sequence demonstrated both a failure mode and its recovery:

> **Turn 1:** User asks 'Which area is most compelling for a deep dive?' System recommends SLC residency control, citing four independent sources by name. The recommendation is well-sourced but wrong: SLC control is a documented dead path in the project.
>
> **Turn 2:** User asks 'Is SLC control a dead path for us?' The narrative retrieval fires (30 records, 12 sessions, arc type: reversal, 39ms). The system reverses its recommendation with the full evidence chain: initial promise, measurement, dead-path declaration.
>
> **Turn 3:** User asks 'What is the next best area, in light of the dead paths?' With seven dead-path records in context, the system recommends an open area (ANE frontier research) and avoids every killed path.

The self-correction between turns 1 and 2 was not the model reasoning better. It was the architecture providing different context. Turn 1 retrieved two memory snippets. Turn 2 retrieved a 30-record narrative arc typed as a reversal. Same model, same weights, same session. The retrieval format changed, and the answer quality changed with it. This sequence also reveals a known limitation: open-ended recommendation queries bypass the contradiction checker, which scans user input but not system output. The system had the dead-path record for SLC but did not check its own recommendation against it. Post-generation entity scanning against the correction store would close this gap.

6.5 Self-Reinforcing Hallucination and Provenance

The most significant failure mode discovered during extended testing was a self-reinforcing hallucination loop. When asked about a topic absent from all memory stores, the 72B generated a detailed fabrication. The 8B extraction model stored this fabrication as typed perceptions with the same authority as user-provided facts. On subsequent queries, the system retrieved its own prior fabrication at score 0.922 and generated answers grounded in it. Each cycle reinforced the original hallucination.

Of 7,640 memories in the production store, 2,668 (35%) had source_role 'assistant'. These model-generated records scored identically to user-provided facts during retrieval. The fix is provenance-weighted retrieval: a 0.5x score multiplier on assistant-sourced records, combined with topical relevance checking that verifies retrieved records address the query's specific terms, not just its entities. Before the fix: a 6-paragraph fabrication about Linux deployment. After: 'This topic hasn\\'t been explored in our research.'

This finding generalizes: any RAG architecture that stores model-generated content alongside user-provided content without provenance weighting is vulnerable to the same feedback loop. Provenance is not metadata. It is an architectural requirement.

7. Related Work

RAG and long-term memory. Retrieval-Augmented Generation (Lewis et al., 2020) and its descendants treat memory as a uniform embedding store queried by similarity. MemGPT (Packer et al., 2023) introduces tiered memory management inspired by operating system virtual memory. Our work differs in arguing that the storage format itself, not the management strategy, determines retrieval quality. Format-matched stores with format-appropriate retrieval and consolidation address failures that no amount of retrieval sophistication can fix when the format is wrong.

Geometric interference in memory. Gopinath (2026) demonstrates that embedding-based memory systems reproduce human forgetting curves due to geometric interference in low-dimensional effective spaces. Our consolidation loops (semantic supersession) directly manage this interference, and our multi-signal retrieval expands effective dimensionality beyond what the embedding model provides. The measurement registry eliminates geometric interference for point-value knowledge by avoiding the embedding space entirely.

Context engineering. Karpathy (2025) frames the core challenge as context engineering: the art of filling the context window with just the right information for the next step. This framing describes the problem without prescribing a solution, and a common response has been to add sophistication inside a single retrieval pipeline (better embeddings, reranking, query rewriting, longer windows). Our contribution is that context engineering is not one problem but several. Different knowledge types fail for different reasons: point values interfere geometrically, narratives fragment when retrieved as isolated snippets, corrections must be checked adversarially rather than ranked, and session state needs recency rather than similarity. A uniform retrieval mechanism cannot serve all of these because the failures are not retrieval failures but format mismatches. Format is the primary design variable; retrieval sophistication is secondary and cannot compensate for the wrong format.

Heterogeneous inference. The ANE dispatch stack builds on the reverse engineering lineage of hollance, maderix, and Kumaresan (Orion, 2025). Our ane-compiler, ane-dispatch, and ane-toolkit repositories extend this work to Q8 quantization, 8B-scale models, and SharedEvents for cross-accelerator coordination. The Subconscious architecture requires this heterogeneous infrastructure because concurrent conscious and subconscious processing demands independent compute paths.

Parallel scan for linear recurrences. Yang et al. (2024) show that the delta-rule recurrence in GatedDeltaNet admits chunked parallel processing via the WY representation (Bischof & Van Loan, 1985), achieving sub-quadratic prefill. Current MLX implementations use the sequential kernel. The 31% sequential compute cost in our GDN-based perception model is an implementation gap, not a mathematical necessity. A Metal port of the chunked WY kernel is future work for the inference stack, separate from the cognitive architecture.

8. Discussion

8.1 LLMs as Shipped Are Structurally Incomplete

The five failure modes we identify are not deficiencies of the model. They are absences in the architecture. A brain without a hippocampus exhibits anterograde amnesia: brilliant in the moment, unable to form new memories. That is the operational profile of every deployed LLM. The industry response has been to make the cortex bigger: more parameters, longer context windows, better training data. This is treating amnesia by making the patient smarter. The patient was never deficient in reasoning. The patient is missing the structures that turn momentary cognition into persistent, formatted, retrievable knowledge.

8.2 Scaling Will Not Fix Format Mismatches

A larger model will not resolve the measurement confusion between DRAM bandwidth and SLC bandwidth when both are presented as similar records in the context window. A larger context window makes the problem worse: more similar records, more interference, more opportunities for the model to select the wrong value. More training data does not help because the measurements are user-specific and post-training. Only storing measurements as measurements, directly addressable by entity name rather than embedded in a similarity space, fixes the measurement confusion. Format is the solution. Scale is not.

8.3 Limitations and Future Work

The evaluation is self-referential: the system was built by the author, trained on the author's data, and evaluated on queries about the author's research. Cross-user and cross-domain validation is necessary to establish generality. Comparison against existing memory-augmented systems (MemGPT, LangChain memory modules) would establish competitive positioning. Evaluation with multiple base models (Llama 70B, Mixtral) would confirm that the improvement is architectural rather than model-specific. These comparisons are future work.

The measurement registry is currently manually curated (39 entries). Auto-population from the extraction pipeline requires broadening the fact-to-registry promotion rules beyond numeric measurements to include all point-value knowledge (names, dates, identifiers). The intent filter uses keyword matching tuned to one user's query patterns. A taxonomy-based classifier using the extraction categories would generalize across users. The post-generation dead-path check (scanning system output against the correction store) would prevent the open-ended recommendation failure demonstrated in the SLC sequence; the frequency of this failure mode under open-ended queries has not been systematically measured. All are concrete engineering tasks with clear specifications.

The longer-term direction is self-architecture: a system that runs its own evaluation harness, detects format mismatches from retrieval quality degradation, and builds new format-matched stores autonomously. The measurement registry was discovered through exactly this process in this work, with a human in the loop. Automating that loop is the path to a system that develops new cognitive capabilities from the user's own data, shaped by the user's own patterns of inquiry, without human intervention.

9. Conclusion

We have presented Subconscious, a continuous cognitive architecture that resolves the five major failure modes of deployed LLMs through format-matched memory. The core finding is that storage format determines retrieval quality: different knowledge types require different storage formats, retrieval mechanisms, and consolidation strategies. A 39-entry lookup table resolved factual recall failures that a full multi-signal retrieval pipeline with 7,395 records could not, because the format matched the retrieval need.

The architecture runs as a continuous background process on heterogeneous Apple Silicon, with conscious generation on GPU and subconscious perception on ANE operating concurrently without measurable contention. The system perceives, integrates, consolidates, and prepares continuously, not in phases. When the user is reading, the subconscious is extracting. When the terminal is closed, the subconscious is consolidating. When the terminal opens, the subconscious has already prepared.

The model did not change. The architecture around it did. The five failure modes of deployed LLMs are not model deficiencies. They are architectural absences. Format-matched memory, running as a continuous cognitive process, fills them.

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Appendix A: Measurement Registry Entries (Sample)

Sample entries from the 39-entry canonical measurement registry demonstrating the entity-value-provenance structure.

Entity	Value	Source Session
SLC geometry	21-way, 128B lines, ~48MB	Main 27
DRAM bandwidth	307 GB/s	Main 33
GPU sustained bandwidth	241 GB/s	Main 33
ANE transport	128 GB/s aggregate (dual NoC)	Main 30
AMCC ceiling	288 GB/s shared	Main 33
8B ANE throughput	7.9 tok/s (Q8)	Main 12
72B warm throughput	29.3 tok/s (spec decode)	Main 25
MiniLM embed latency	0.84ms	Main 14
Neuron routing latency	905 microseconds	Main 14
ANE SRAM cliff	32MB (4x latency above)	Main 28

Appendix B: Repository Index

Repository	Description	License
github.com/MidasMulli/subconscious	Cognitive memory system	MIT
github.com/MidasMulli/ane-compiler	ANE compilation pipeline (no CoreML)	MIT
github.com/MidasMulli/ane-dispatch	Direct _ANEClient dispatch runtime	MIT
github.com/MidasMulli/ane-toolkit	ANE binary format specification	MIT
github.com/MidasMulli/apple-slc-cache-hints	SLC geometry + measurement tools	MIT
github.com/MidasMulli/orion-ane	Midas cognitive agent	MIT